



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

TG: @Chalnaayaaar NEET (UG)

MINOR

16-07-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E5

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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Your Target is to secure Good Rank in Pre-Medical 2024

Topic : WAVE OPTICS, THERMAL PHYSICS, UNIT AND DIMENSION, CURRENT ELECTRICITY, CAPACITOR, MEC (TILL DATE) [BACK UNIT] : CURRENT ELECTRICITY

SECTION - A (PHYSICS)

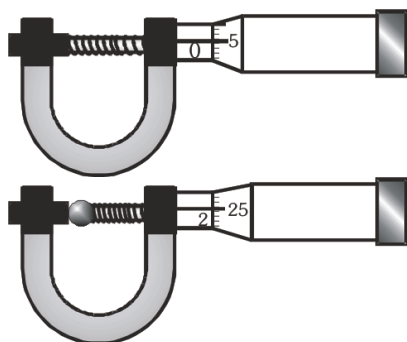
1. In an experiment four quantities a,b,c and d are measured with percentage error 0.25%, 0.5%, 1% and 1.5% respectively. Quantity P is calculated as follows $P = \frac{a^4 b^3}{c^2 d}$ % error in p is:

(1) 4% (2) 5% (3) 6% (4) 7%

2. If the energy (E), velocity (v) and force (F) be taken as the fundamental quantity, then the dimensions of mass will be :-

(1) Fv^{-2} (2) Fv^{-1} (3) Ev^{-2} (4) Ev^2

3. The number of circular division on the shown screw gauge is 100. It moves 1 mm on main scale for one complete rotation and main scale has 1 mm marks. The diameter of the ball is :-



(1) 2.25 mm (2) 2.20 mm
(3) 1.20 mm (4) 1.25 mm

4. Write down the number of significant figure in 0.04320 N/m^2

(1) 5 (2) 3 (3) 2 (4) 4

5. In Young's double slit experiment the y-co-ordinates of central maximum and 10th maxima are 2 cm and 5 cm respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5, the corresponding y-co-ordinates will be :-

(1) 2 cm, 7.5 cm (2) 3 cm, 6 cm
(3) 2 cm, 4 cm (4) $\frac{4}{3}$ cm, $\frac{10}{3}$ cm

6. Young's Double Slit Experiment is first performed in air then in a medium other than air. It is found that 10th bright fringe in medium lies where 8th dark fringe lies in air. The refractive index of medium is nearly :-

(1) 1.59 (2) 1.78 (3) 1.69 (4) 1.33

7. The central fringe of the interference pattern produced by light of wavelength $6000 \text{ \AA}$ is found to shift to the position of 4th bright fringes after a glass plate of refractive index 1.5 is introduced. The thickness of the glass plate would be :-

(1) $4.8 \mu\text{m}$ (2) $8.23 \mu\text{m}$
(3) $14.98 \mu\text{m}$ (4) $3.78 \mu\text{m}$

8. A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aperture is illuminated by a parallel beam of wavelength $5 \times 10^{-5} \text{ cm}$. The distance of first dark band of diffraction pattern from the centre of screen-

(1) 0.20 cm (2) 0.15 cm
(3) 0.10 cm (4) 0.25

9. What should be the width of each slit be to obtain 20 maxima of double slit pattern within the central maximum of the single slit pattern (distance between both slit = 2 mm)

(1) 0.2 mm (2) 2 mm
(3) 4 mm (4) 0.6 mm

10. A beam of natural light falls on a system of 4 polaroids, which are arranged in succession such that pass axis of each polaroid is turned through 60° with respect to the preceding one. The fraction of the incident light intensity that passes through the system is-

(1) $\frac{1}{64}$ (2) $\frac{1}{32}$ (3) $\frac{1}{512}$ (4) $\frac{1}{128}$

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PHASE - MED

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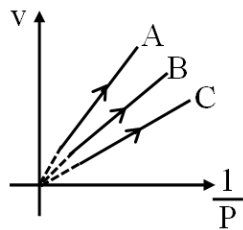
11. Light waves can be polarized because they-

- (1) have high frequency
- (2) have short wavelength
- (3) are transverse
- (4) can be reflected

12. A vessel has 6g of oxygen at pressure P and temperature 400 K. A small hole is made in it so that oxygen leaks out. How much oxygen leaks out if the final pressure is $P/2$ and temperature is 300 K?

- (1) 3g (2) 2g (3) 4g (4) 5g

13. Figure shows isotherms of fixed mass of an ideal gas at three temperatures T_A , T_B and T_C then :-

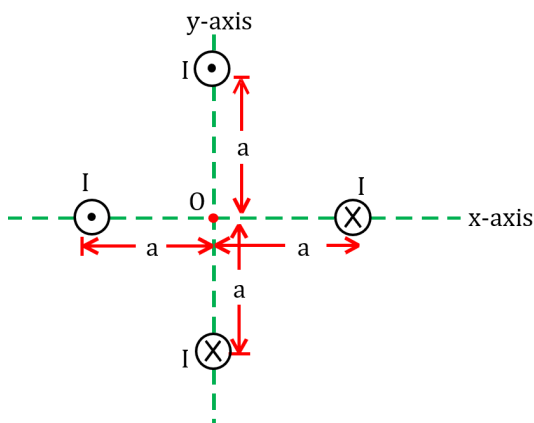


- (1) $T_A > T_B > T_C$ (2) $T_A < T_B < T_C$
- (3) $T_B > T_A > T_C$ (4) $T_A = T_B = T_C$

14. A long straight wire in the horizontal plane carries a current in north to south direction. Give direction of B at a point east of the wire :-

- (1) Upwards (2) Downwards
- (3) North (4) West

15. For given current distribution, each infinite length wire produces magnetic field ' B ' at origin then resultant magnetic field at origin 'O' is :-



- (1) $4B$ (2) $\sqrt{2} B$ (3) $2\sqrt{2} B$ (4) zero

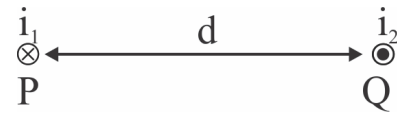
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PHASE - MED

16. Same current I is flowing in three infinitely long wire along positive x , y and z -directions. The magnetic field at a point $(0, 0, -a)$ would be :-

- (1) $\frac{\mu_0 I}{2\pi a}(\hat{j} - \hat{i})$
- (2) $\frac{\mu_0 I}{2\pi a}(\hat{i} + \hat{j})$
- (3) $\frac{\mu_0 I}{2\pi a}(\hat{i} - \hat{j})$
- (4) $\frac{\mu_0 I}{2\pi a}(\hat{i} + \hat{j} + \hat{k})$

17. Two parallel wires P and Q placed at a separation $d = 6$ cm carry electric currents $i_1 = 5$ A and $i_2 = 2$ A in opposite directions as shown in figure. Find the point on the line PQ where the resultant magnetic field is zero :



- (1) 4 cm from Q
- (2) 2 cm from Q
- (3) 5 cm from P
- (4) 3 cm from P

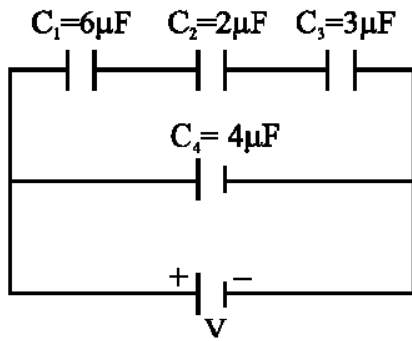
18. If the magnetic flux is expressed in weber, then the magnetic induction can be expressed in :

- (1) weber/m² (2) weber/m
- (3) weber-m (4) weber-m²

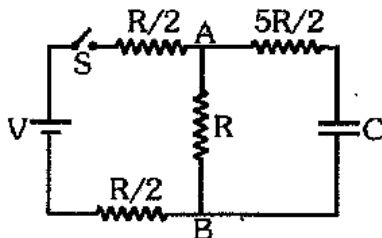
19. The vector form of Biot savart law for a current carrying element is :-

- (1) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\ell \sin \phi}{r^2}$
- (2) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{\ell} \times \vec{r}}{r^2}$
- (3) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{\ell} \times \hat{r}}{r^3}$
- (4) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{id\vec{\ell} \times \vec{r}}{r^2}$

20. In the given circuit diagram, the ratio of the charges on capacitors C_2 and C_4 is:-



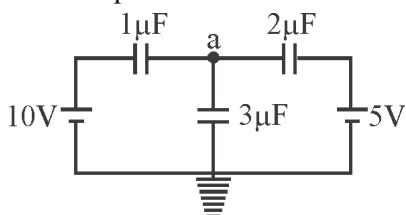
- (1) 4 (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) 2
21. In the circuit shown in figure, the battery is an ideal one with emf V . The capacitor is initially uncharged. Switch S is closed at time $t = 0$.



The final charge Q on the capacitor is :-

- (1) $\frac{CV}{2}$ (2) $\frac{CV}{3}$ (3) CV (4) $\frac{CV}{6}$
22. A battery charges a parallel plate capacitor of thickness (d) so that an energy $[U_0]$ is stored in the system. A slab of dielectric constant (K) and thickness (d) is introduced between the plates (Keeping battery connected) of the capacitor. The new energy of the system is given by :-

- (1) KU_0 (2) K^2U_0 (3) $\frac{U_0}{K}$ (4) $\frac{U_0}{K^2}$
23. Find potential at point 'a' :-



- (1) $\frac{20}{3}$ volt (2) 5 volt
(3) $10/3$ volt (4) $3/20$ volt

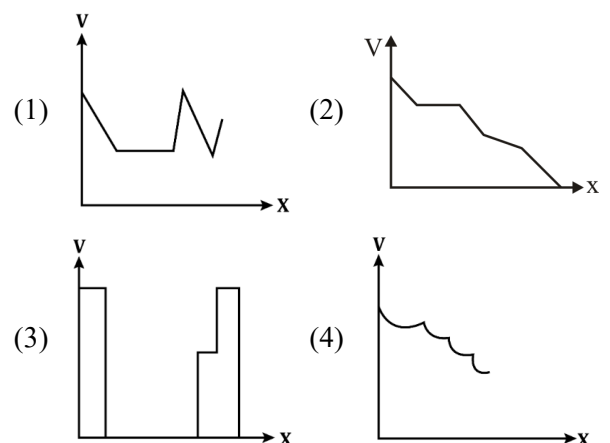
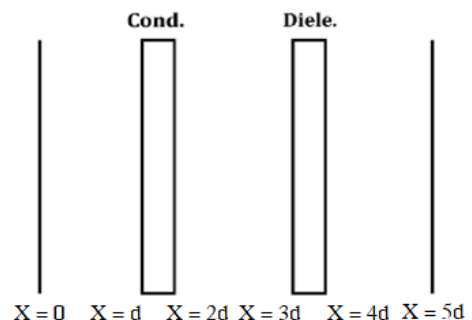
24. The energy required to charge a parallel plate condenser of plate separation d and plate area of cross-section A such that the uniform electric field between the plates is E , is :-

- (1) $\frac{1}{2} \epsilon_0 E^2 / Ad$ (2) $\epsilon_0 E^2 / Ad$
(3) $\epsilon_0 E^2 Ad$ (4) $\frac{1}{2} \epsilon_0 E^2 Ad$

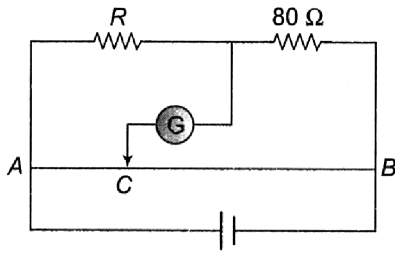
25. A parallel plate air capacitor is charged to a potential difference of V volts. After disconnecting the charging battery the distance between the plates of the capacitor is increased using an insulating handle. As a result the potential difference between the plates

- (1) does not change (2) becomes zero
(3) increases (4) decreases

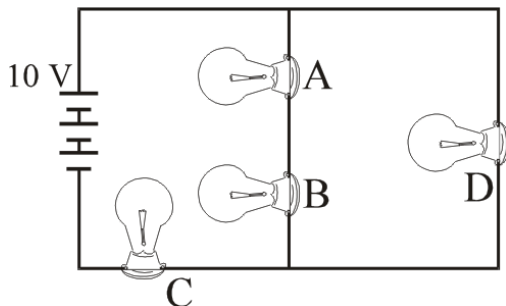
26. The distance between plates of a parallel plate capacitor is $5d$. The positively charged plate is at $X = 0$ and negatively charged plates is at $X = 5d$. Two slabs - one of conductor and the other of a dielectric, both of same thickness d are inserted between the plates as shown in figure. Potential (V) versus distance x graph will be (consider potential of negatively charged plate to be zero)



27. AB is a wire of uniform resistance. The galvanomere G shows no current when the length $AC = 20$ cm and $CB = 80$ cm. The resistance R is equal to :



- (1) 2Ω (2) 8Ω
 (3) 20Ω (4) 40Ω
28. A galvanometer of resistance 100Ω contains 100 division. It gives a deflection of one division on passing a current of 10^{-4} A. Find the resistance in ohms to be connected to it, so that it becomes a voltmeter of range 10V.
- (1) $\frac{500}{9} \Omega$ (2) 500Ω
 (3) $\frac{100}{9} \Omega$ (4) 900Ω
29. In a potentiometer experiment, the balancing with a cell is at length 240 cm. On shunting the cell with a resistance of 2Ω the balancing length becomes 120 cm. The internal resistance of the cell is :
- (1) 2Ω (2) 4Ω
 (3) 0.5 (4) 1Ω
30. In the circuit diagram, all the bulbs are identical. Which bulb will be the brightest?



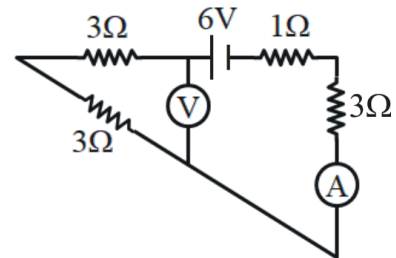
- (1) A (2) B
 (3) C (4) D

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31. **Statement-I** : The capacity of a conductor, under given circumstance, remain constant irrespective of the charge present on it.

Statement-II : Capacity depends on size and shape of conductor and also on the medium.

- (1) Statement-I and II both are correct.
 (2) Statement-I and II both are incorrect.
 (3) Only Statement-I is correct
 (4) Only Statement-II is correct
32. In the given circuit ammeter and voltmeter are ideal and battery of 6V has internal resistance 1Ω .



Which of the following statement are correct :-

- (A) Current through battery is 0.6 A
 (B) Reading of Ammeter is 1A
 (C) Reading of voltmeter is 3.6 V
 (D) Terminal potential difference of cell is 5V.
- (1) B,D (2) A,C
 (3) A,C,D (4) B,C,D
33. A long straight wire in the horizontal plane carries a current of 50A in North to South direction. Give the magnitude and direction of B at a point 2.5 m west of the wire :
- (1) 2×10^{-6} T, South (2) 3×10^{-6} T, Up
 (3) 4×10^{-6} T, Down (4) 6T, North
34. A parallel plate capacitor with air between the plates has a capacitance of 8 pF. What will be the capacitance if the distance between the plates is reduced by half and the space between them is filled with a substance of dielectric constant 6 ?
- (1) 24 pF (2) 96 pF
 (3) 8 pF (4) 12 pF

35. Three capacitors each of capacitance 9 pF are connected in series. What is the potential difference across each capacitor, if the combination is connected to a 120V supply.

- (1) 40V (2) 60V
(3) 80V (4) 50V

SECTION - B (PHYSICS)

36. A vernier callipers has 20 divisions on the vernier scale which coincide with 19 divisions on the main scale. The least count of the instrument is 0.2 mm. Then find out the value of main scale division :-

- (1) 0.5 mm
(2) 4 mm
(3) 2 mm
(4) $\frac{1}{4}$ mm

37. Light of wavelength 6000 Å is incident on a thin glass plate of refractive index 1.5 such that angle of refraction into the plate is 60° calculate the smallest thickness of the plate which will make it appear dark by reflection :-

- (1) 4×10^{-6} m
(2) 4×10^{-7} m
(3) 8×10^{-6} m
(4) 8×10^{-7} m

38. It is found that when two waves of same intensity from two coherent sources superimpose at a certain point, then the resultant intensity is equal to the intensity of any one wave. This means that the phase difference between the two waves at that point is-

- (1) 0 (2) $\frac{\pi}{2}$
(3) $\frac{2\pi}{3}$ (4) π

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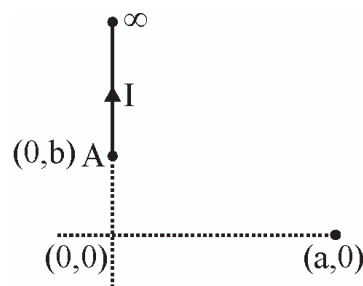
39. In Young's double slit using red and blue lights of wavelength 600 nm and 480 nm respectively the value of n for which n^{th} red fringe coincides with $(n + 1)^{\text{th}}$ blue fringe is :

- (1) 5
(2) 4
(3) 3
(4) 2

40. Two polaroids are placed at 37° to each other. The percentage of incident light passing through combination :-

- (1) 8%
(2) 16%
(3) 32%
(4) 64%

41. An infinitely long wire carrying current I is along Y axis such that its one end is at point A(0, b) while the wire extends upto $+\infty$. The magnitude of magnetic field strength at point (a, 0) is :-



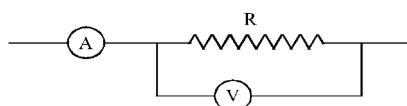
- (1) $\frac{\mu_0 I}{4\pi a} \left(1 + \frac{b}{\sqrt{a^2 + b^2}} \right)$
(2) $\frac{\mu_0 I}{4\pi a} \left(1 - \frac{b}{\sqrt{a^2 + b^2}} \right)$
(3) $\frac{\mu_0 I}{4\pi a} \left(\frac{b}{\sqrt{a^2 + b^2}} \right)$
(4) None of these

42.

List-I [Orientation of current carrying conductor]		List-II [Value of magnetic field at point P]	
(P)		(A)	$\frac{\mu_0 I}{4\pi R} \left(\frac{3\pi}{2} + 2 \right) \odot$
(Q)		(B)	$\frac{\mu_0 I}{4\pi R} \left(\frac{3\pi}{2} - 2 \right) \odot$
(R)		(C)	$\frac{\mu_0 I}{4R} \otimes$
(S)		(D)	$\frac{3\mu_0 I}{8R} \odot$

	P	Q	R	S
(1)	C	D	B	A
(2)	A	D	B	C
(3)	D	B	A	A
(4)	C	D	A	B

43. In the adjoining circuit diagram, the readings of ammeter and voltmeter are 2A and 120V respectively. If the value of R is 75Ω then the voltmeter resistance will be.



- (1) 100Ω (2) 150Ω (3) 300Ω (4) 75Ω

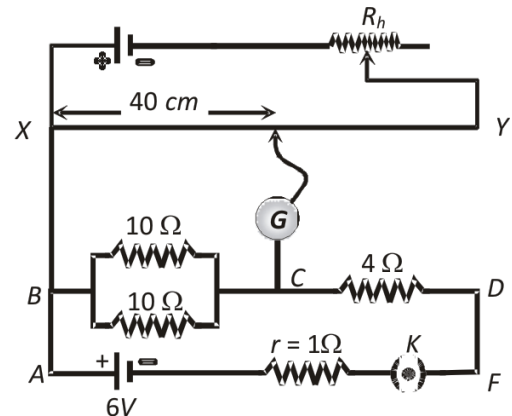
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PHASE - MED

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44.

In the following circuit the potential difference between the points B and C is balanced against 40 cm length of potentiometer wire. In order to balance the potential difference between the points C and D, where should jockey be pressed



- (1) 32 cm (2) 16 cm
(3) 8 cm (4) 4 cm

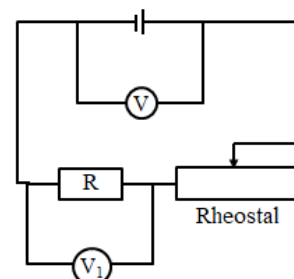
45.

An electric bulb of rating 500 W, 100 V is used in a circuit having a 200 V supply. The resistance R that must be used with the bulb, so that the bulb uses 500 W is :-

- (1) 10Ω , in series
(2) 10Ω , in parallel
(3) 20Ω , in series
(4) 20Ω , in parallel

46.

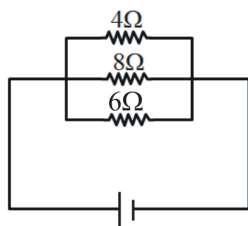
If the rheostat slider were to move from extreme right to the far left. How will the reading of voltmeter V_1 change?



- (1) First increase and then decrease
(2) First decrease and then increase
(3) Increase continuously
(4) Decrease continuously

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47. In the circuit shown the total power developed in the 4Ω and 8Ω resistor is 18 W.

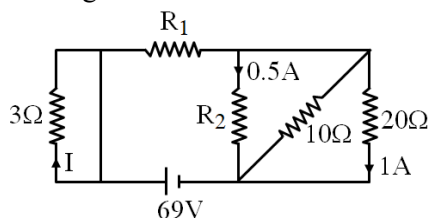


Based on above information match Column-I with Column-II.

	Column-I		Column-II
A.	Power dissipated in 4Ω is	P.	6W
B.	Power dissipated in 8Ω is	Q.	10W
C.	Total power dissipated in 6Ω & 4Ω is	R.	12W
D.	Total power dissipated in 8Ω & 6Ω is	S.	14W
E.	Power dissipated in 6Ω is	T.	18W
		U.	8W

- (1) (A–R), (B–U), (C–Q), (D–T), (E–P)
 (2) (A–Q), (B–U), (C–R), (D–T), (E–P)
 (3) (A–R), (B–P), (C–T), (D–S), (E–U)
 (4) (A–Q), (B–P), (C–T), (D–S), (E–U)

48. Consider the given circuit-

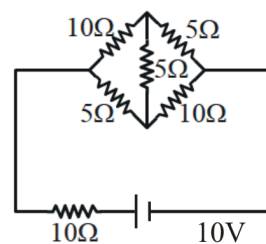


For above circuit match the column :

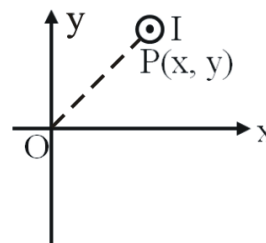
	Column-I		Column-II
A.	I (in ampere)	P.	30
B.	R_1 (in Ohm)	Q.	40
C.	Power dissipated in 10Ω resistor (in Watt)	R.	14
D.	R_2 (in Ohm)	S.	3
		T.	0

- (1) (A–S), (B–R), (C–P), (D–Q)
 (2) (A–S), (B–P), (C–Q), (D–R)
 (3) (A–T), (B–R), (C–Q), (D–Q)
 (4) (A–T), (B–R), (C–Q), (D–P)

49. Determine the current drawn from the source-



- (1) $\frac{4}{17}$ A
 (2) $\frac{10}{17}$ A
 (3) $\frac{6}{17}$ A
 (4) $\frac{5}{17}$ A
50. A current carrying infinitely long straight wire is placed at point P as shown, the magnetic field due to wire at origin is-



- (1) $B = \frac{\mu_0 I}{2\pi(x^2 + y^2)} [\hat{y}\hat{i} - \hat{x}\hat{j}]$
 (2) $B = \frac{\mu_0 I}{2\pi(x^2 + y^2)} [-\hat{y}\hat{i} + \hat{x}\hat{j}]$
 (3) $B = \frac{\mu_0 I}{2\pi\sqrt{x^2 + y^2}} [\hat{i} - \hat{j}]$
 (4) $B = \frac{\mu_0 I}{2\pi(x^2 + y^2)^{3/2}} [+y\hat{i} + x\hat{j}]$

Topic : NSR, ELIMINATION REACTION, HYDROCARBON, OXIDATION, PERIODIC TABLE
(COMPLETE) [BACK UNIT] : ELECTROCHEMISTRY

SECTION-A (CHEMISTRY)

51. Which atomic number is transuranium element :

- (1) 101 (2) 92
(3) 118 (4) (1) and (3) both

52. Which of the following has the highest value of ΔH_{eg} :-

- (1) F (2) Cl
(3) C (4) Ne

53. Successive ionisation energies of an element 'X' are given below (in K. Cal)

IP ₁	IP ₂	IP ₃	IP ₄
165	195	556	595

Electronic configuration of the element 'X' is :-

- (1) $1s^2, 2s^2 2p^6, 3s^2 3p^2$
(2) $1s^2, 2s^1$
(3) $1s^2, 2s^2 2p^2$
(4) $1s^2, 2s^2 2p^6, 3s^2$

54. The relative penetration power of different subshells is :

- (1) $s > p > d > f$ (2) $s < p < d < f$
(3) $s < d < p < f$ (4) $d < s < p < f$

55. $X = [Ar]3d^{10}4s^1$ belongs from :

- (1) 2 group (2) 11 group
(3) 12 group (4) 14 group

56. The first ionisation energy (in kJ/mol) of Na, Mg, Al and Si respectively are :

- (1) 496, 577, 737, 786
(2) 786, 737, 577, 496
(3) 496, 577, 786, 737
(4) 496, 737, 577, 786

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57. Match the column I with column II and choose the correct option using the codes given below

	Column-I (Elements)		Column II (IUPAC name)
(A)	109	(1)	Ununbium
(B)	112	(2)	Unnilennium
(C)	115	(3)	Ununpentium
(D)	118	(4)	Ununoctium

- (1) A-1, B-2, C-3, D-4 (2) A-2, B-1, C-3, D-4
(3) A-1, B-2, C-4, D-3 (4) A-2, B-1, C-4, D-3

58. In 5th period of long form of periodic table, electronic energy levels are in the order :-

- (1) 5s, 5p, 4d (2) 5s, 4d, 5p
(3) 5s, 4f, 3d, 4p (4) 6s, 4f, 5d, 4p

59. From atomic number 90 to 103, element are placed in :-

- (1) 6th period, III A group
(2) 5th period, III B group
(3) 6th period, III B group
(4) 7th period, III B group

60. Which of the following has largest radius :-

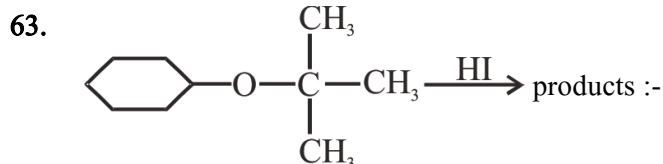
- (1) $1s^2, 2s^2, 2p^6, 3s^2 3p^5$
(2) $1s^2, 2s^2, 2p^6, 3s^1$
(3) $1s^2, 2s^2, 2p^6, 3p^3$
(4) $1s^2, 2s^2, 2p^6, 3p^4$

61. Which of the following is different from other :

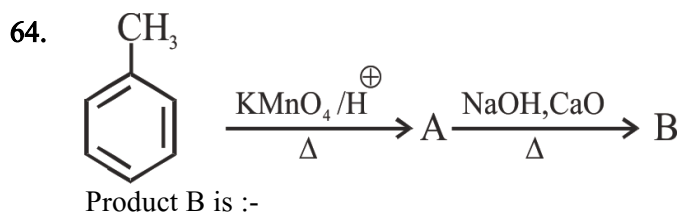
- (1) ZnO (2) BeO (3) Al_2O_3 (4) K_2O

62. Which of the following order of acidic strength is correct?

- (1) $H_2S < H_2Se < H_2Te$
(2) $NH_3 < PH_3 < AsH_3$
(3) $HF < HCl < HBr < HI$
(4) All



- (1) Cyclohexyl-OH and $\text{CH}_3\text{-C(CH}_3)_2\text{-I}$
- (2) Cyclohexyl-I and $\text{CH}_3\text{-C(CH}_3)_2\text{-OH}$
- (3) $\text{Cyclohexyl-O}^\ominus$ and $\text{CH}_3\text{-C(CH}_3)_2^\oplus$
- (4) I-Cyclohexyl-OH and $\text{CH}_3\text{-C(CH}_3)_2\text{-CH=CH}_2$



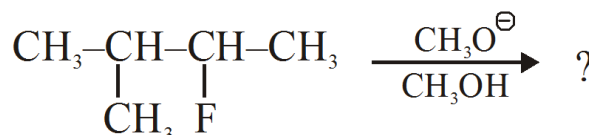
- (1) C_6H_6 (2) $\text{C}_6\text{H}_5\text{OH}$
- (3) $\text{C}_6\text{H}_5\text{COOH}$ (4) $\text{C}_6\text{H}_5\text{CHO}$

65. Which of the following is most reactive towards $\text{S}_\text{N}1$ reaction ?

- (1) $\text{p-NO}_2\text{-C}_6\text{H}_4\text{-CH}_2\text{-Br}$ (2) $\text{p-CH}_3\text{-C}_6\text{H}_4\text{-CH}_2\text{-Br}$
- (3) $\text{p-OH-C}_6\text{H}_4\text{-CH}_2\text{-Br}$ (4) $\text{p-Cl-C}_6\text{H}_4\text{-CH}_2\text{-Br}$

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66. Identify the product ?



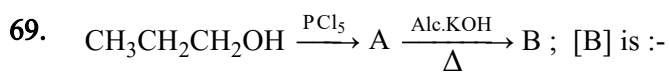
- (1) $\text{CH}_3\text{-CH(CH}_3\text{)-CH(OCH}_3\text{)-CH}_3$
- (2) $\text{CH}_3\text{-CH(CH}_3\text{)-CH=CH}_2$
- (3) $\text{CH}_3\text{-C(CH}_3\text{)=CH-CH}_3$
- (4) $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$

67. Which of the following is most reactive halide in Williamson ether synthesis ?

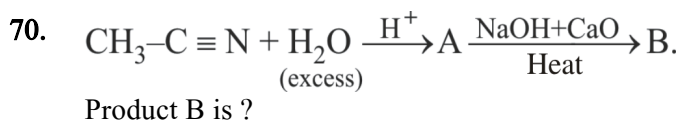
- (1) $\text{CH}_3\text{-Cl}$ (2) $\text{CH}_3\text{-CH(CH}_3\text{)-Cl}$
- (3) $\text{CH}_3\text{-C(CH}_3)_2\text{-Cl}$ (4) $\text{CH}_3\text{-C(CH}_3)_2\text{-CH}_2\text{-Cl}$

68. Most reactive toward acid catalyst dehydration is:-

- (1) $\text{1,2-dihydronaphthalen-1-ol}$ (2) $\text{1,2-dihydronaphthalen-2-ol}$
- (3) $\text{1,4-dihydronaphthalen-1-ol}$ (4) $\text{1,4-dihydronaphthalen-2-ol}$



- (1) Propene (2) Propyne
- (3) Propanone (4) Propanal

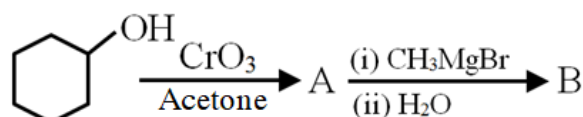


- (1) $\text{CH}_3\text{-C(=O)-NH}_2$ (2) $\text{CH}_3\text{-C(=O)-OH}$
- (3) $\text{CH}_3\text{-C(=O)-ONa}$ (4) CH_4

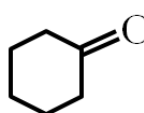
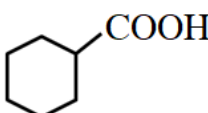
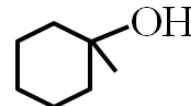
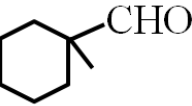
71. Which of the following reaction does not give ethane as product ?

- (1) $\text{CH}_3\text{-Cl} \xrightarrow[\text{dry ether}]{\text{Na}}$
- (2) $\text{CH}_3\text{-COOH} \xrightarrow[\text{lime}]{\text{soda}}$
- (3) $\text{CH}_3\text{-COOK} \xrightarrow{\text{electrolysis}}$
- (4) $\text{CH}_3\text{-CH}_2\text{-O-CH}_2\text{-CH}_3 \xrightarrow[\text{HI}]{\text{Red P}}$

72.



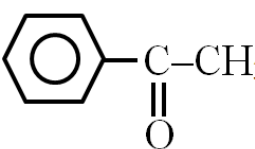
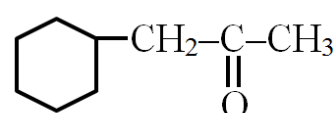
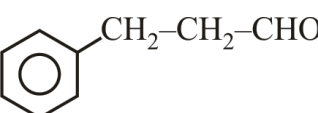
Product B is :

- (1)  (2) 
- (3)  (4) 

73. Anisole can be prepared by the action of methyl iodide on sodium phenoxide. The reaction is called :

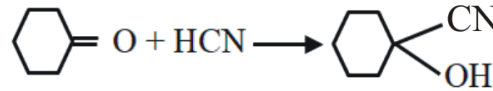
- (1) Fittig reaction
- (2) Kolbe electrolysis
- (3) Williamson ether synthesis
- (4) Wurtz reaction

74. Which of the following compound will not give yellow ppt. with $\text{I}_2 / \text{OH}^\ominus$?

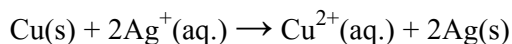
- (1) 
- (2) 
- (3) 
- (4) $\text{CH}_3\text{-CHO}$

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75. Which of the following is E_1 elimination ?

- (1) $\text{CH}_3\text{-CH=CH}_2 \xrightarrow{\text{HBr}} \text{CH}_3\text{-CH(Br)-CH}_3$
- (2) $\text{CH}_3\text{-C(CH}_3)_2\text{-OH} \xrightarrow[\Delta]{\text{Conc. H}_2\text{SO}_4} \text{CH}_3\text{-C(CH}_3)=\text{CH}_2$
- (3) 
- (4) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3 \xrightarrow[\text{light}]{\text{Sun}} \text{CH}_3\text{-CH}_2\text{-CH(Cl)-CH}_3$

76. For the cell reaction :



- (1) $E_{\text{cell}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{Ag}^+/\text{Ag}}$
- (2) $E_{\text{cell}} = E_{\text{Ag}^+/\text{Ag}} - E_{\text{Cu}^{2+}/\text{Cu}}$
- (3) $E_{\text{cell}} = E_{\text{Ag}^+/\text{Ag}} + E_{\text{Cu}^{2+}/\text{Cu}}$
- (4) $E_{\text{cell}} = 2E_{\text{Ag}^+/\text{Ag}} + E_{\text{Cu}^{2+}/\text{Cu}}$

77. The oxidation potential of hydrogen electrode which is in contact with a solution having pH = 10 is

- (1) 0.591 V (2) 0.295 V
- (3) 1.182 V (4) 0.886 V

78. Among the following incorrect statement is

- (1) For strong electrolytes, Λ_m increases slowly with dilution
- (2) For strong electrolytes, conductivity decreases with decrease in concentration
- (3) For weak electrolytes, Λ_m increases steeply with dilution especially near lower concentration
- (4) For weak electrolytes, conductivity increases with decrease in concentration

79. In the equation : $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$, the value of A will be same for

- (1) NaCl and KCl (2) NaCl and CaCl_2
- (3) CaCl_2 and MgSO_4 (4) KCl and MgSO_4

80. The mass of Ni deposited at cathode by passing 5A current for 20 min through $\text{Ni}(\text{NO}_3)_2$ solution using Pt-electrodes will be (Atomic mass of Ni = 58.7)

- (1) 1.56 g (2) 2.46 g
(3) 1.23 g (4) 1.82 g

81. A steady current of 1.5 ampere was passed through two electrolytic cells A and B containing CuSO_4 and AgNO_3 respectively and connected in series. If 1.45 g of Ag is deposited at the cathode of cell B then the mass of copper deposited in cell A will be

- (1) 0.43 g (2) 0.36 g
(3) 0.51 g (4) 0.59 g

82. A spoon to be electroplated with gold should be made :

- (1) Cathode (2) Anode
(3) Electrolyte (4) All of these

83. The standard reduction potential for half reactions for four different elements. A, B, C and D are :

- (i) $\text{A}_2 + 2\text{e}^- \rightarrow 2\text{A}^-$ $E^\circ = +2.85 \text{ V}$
(ii) $\text{B}_2 + 2\text{e}^- \rightarrow 2\text{B}^-$ $E^\circ = +1.36 \text{ V}$
(iii) $\text{C}_2 + 2\text{e}^- \rightarrow 2\text{C}^-$ $E^\circ = +1.06 \text{ V}$
(iv) $\text{D}_2 + 2\text{e}^- \rightarrow 2\text{D}^-$ $E^\circ = +0.53 \text{ V}$

The strongest oxidising and strongest reducing agents among these :

- (1) would be A_2 and D^- respectively
(2) would be D_2 and A^- respectively
(3) would be B_2 and C^- respectively
(4) cannot be ascertained from the given data as the species being subjected to oxidation or reduction have not been indicated

84. Standard electrode potential of an electrode is :

- (1) Intensive property
(2) Extensive property
(3) Isothermal property
(4) Isobaric property

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85. The molar conductivity of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. The molar conductivity of CH_3COOH at infinite dilution is:

- (1) $201.28 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
(3) $390.71 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $540.45 \text{ S cm}^2 \text{ mol}^{-1}$

SECTION-B (CHEMISTRY)

86. The correct order of decreasing first ionisation energy is :-

- (1) $\text{P} > \text{Cl} > \text{Si} > \text{Al}$ (2) $\text{Cl} > \text{P} > \text{S} > \text{Al}$
(3) $\text{F} > \text{N} > \text{O} > \text{B}$ (4) 2 & 3 both

87. In which of the following process energy will be absorbed ?

- (1) $\text{Na}^+_{(\text{g})} \rightarrow \text{Na}_{(\text{g})}$ (2) $\text{Be}^-_{(\text{g})} \rightarrow \text{Be}_{(\text{g})}$
(3) $\text{O}^-_{(\text{g})} \rightarrow \text{O}^{2-}_{(\text{g})}$ (4) $\text{F}_{(\text{g})} \rightarrow \text{F}^-_{(\text{g})}$

88. Which of the following oxides is most acidic in nature :-

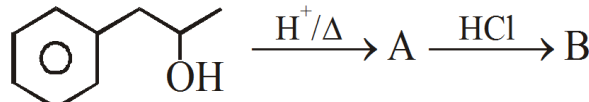
- (1) MgO (2) K_2O (3) ZnO (4) CaO

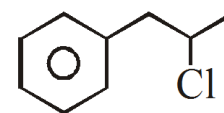
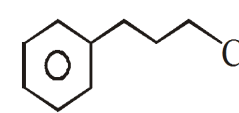
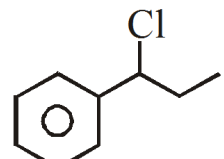
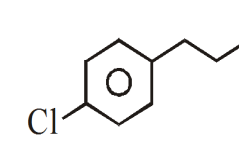
89. The element expected to form largest ion to achieve the nearest noble gas configuration is :-

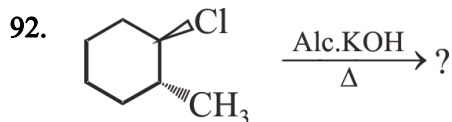
- (1) Cl (2) P (3) Na (4) S

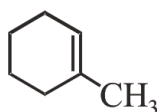
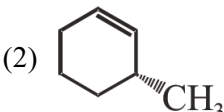
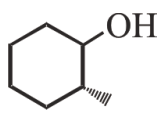
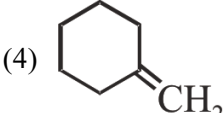
90. Which of the following order electron gain enthalpy is correct?

- (1) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (2) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
(3) $\text{Cl} > \text{F} > \text{S} > \text{O}$ (4) 2 & 3 both

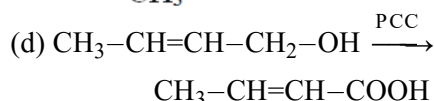
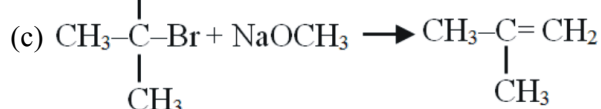
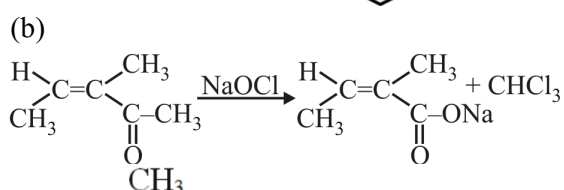
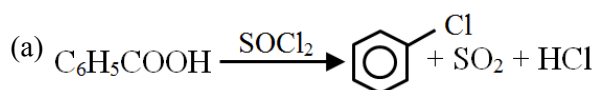
91. 
The product 'B' is :

- (1)  (2) 
(3)  (4) 



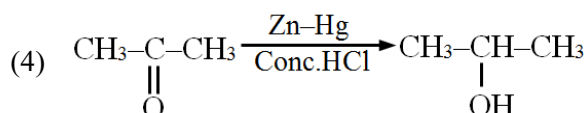
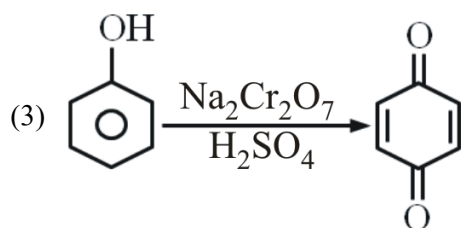
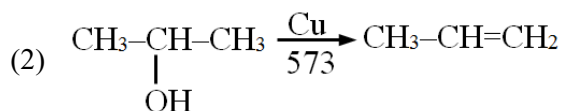
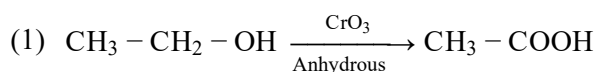
- (1)  (2) 
 (3)  (4) 

93. Which of the following conversion is incorrect ?



- (1) a,d
 (2) b,c
 (3) only d
 (4) only c

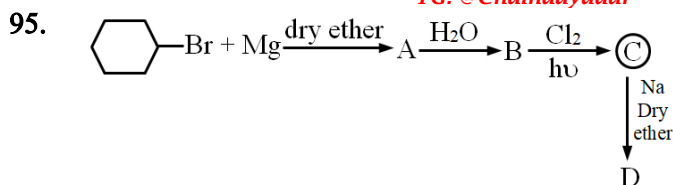
94. Which of the following is correct ?



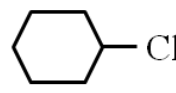
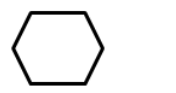
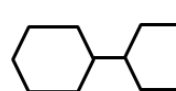
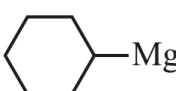
TG: @Chalnaayaaar

PHASE - MED

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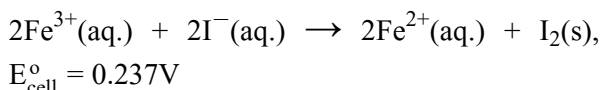
Product D is :

- (1)  (2) 
 (3)  (4) 

96. If an external opposite potential is applied to Daniel cell (in standard state) the reaction continues to take place till the opposite voltage reaches the value of

- (1) 1.1 V (2) 1.8 V (3) 2.1 V (4) 2.3 V

97. The equilibrium constant of the given reaction at 298 K will be



- (1) $10^{12.25}$ (2) $10^{8.02}$ (3) $10^{6.8}$ (4) $10^{9.75}$

98. What products are formed during the electrolysis of concentrated aqueous solution of sodium chloride ?

- (I) $\text{Cl}_2(\text{g})$ at anode
 (II) NaOH as electrolyte
 (III) $\text{H}_2(\text{g})$ at cathode

- (1) I only (2) I and II only
 (3) I and III only (4) I, II and III

99. Resistance of 0.2 M solution of an electrolyte is 50Ω . The specific conductance of the solution is 1.4 S m^{-1} . The resistance of 0.5 M solution of the same electrolyte is 280Ω . The molar conductivity of 0.5 M solution of the electrolyte in $\text{S m}^2\text{ mol}^{-1}$ is :

- (1) 5×10^{-4} (2) 5×10^{-3}
 (3) 5×10^3 (4) 5×10^2

100. $\text{Pt} | \text{H}_2(\text{P}_1\text{ atm}) | \text{H}^+(1\text{M}) || \text{H}^+(1\text{M}) | \text{H}_2(\text{P}_2\text{ atm}) | \text{Pt}$
 P_1 and P_2 are pressure, cell reaction will be spontaneous if :

- (1) $\text{P}_1 = \text{P}_2$ (2) $\text{P}_1 > \text{P}_2$
 (3) $\text{P}_2 > \text{P}_1$ (4) $\text{P}_1 = 1\text{ atm}$

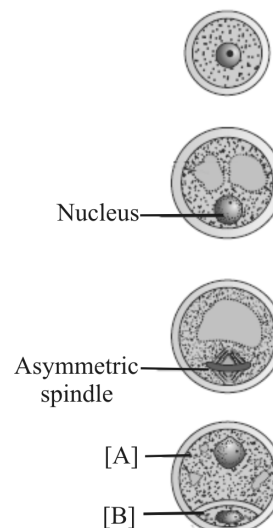
16-07-2023

Topic : DIGESTION AND ABSORPTION, MOLECULAR BASIS OF INHERITANCE UPTO TRANSLATION , TAXONOMY TO BIOLOGICAL CLASSIFICATION AND MONERA [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM

SECTION-A (BIOLOGY-I)

101. In over 40% of angiosperms, pollen grains are shed at 3 celled stage, these 3 cells are :-
- (1) Vegetative cell, tube cell and generative cell
 - (2) One generative cell and two tube cells
 - (3) One vegetative cell and two male gametes
 - (4) One generative cell and two male gametes
102. Which of following layer helps in dehiscence of anther for liberation of pollen grains?
- (1) Epidermis
 - (2) Endothecium
 - (3) Middle layer
 - (4) Endothelium
103. _____ that comes to india as a contaminant with imported wheat, has become ubiquitous in occurance and causes pollen allergy :-
- (1) Parthenium
 - (2) *Triticum aestivum*
 - (3) *Cynodon dactylon*
 - (4) *Amorphophallus*
104. The most developed male gametophyte is found in-
- (1) Mango
 - (2) Cycas
 - (3) Pinus
 - (4) Sphagnum
105. How many meiotic division/divisions is/are required for the formation of eight male gametes in a typical angiosperm ?
- (1) One
 - (2) Two
 - (3) Three
 - (4) Four

106. Stages of a microspore maturing into a pollen grain are given in the diagrams. Identify labelled cells [A] and [B] of given diagram :-



- (1) [A] Generative cell, [B] Vegetative cell
 - (2) [A] Generative cell, [B] Tube cell
 - (3) [A] Vegetative cell, [B] Tube cell
 - (4) [A] Vegetative cell, [B] Generative cell
107. If leaf tip cells posses 28 chromosomes, the number of chromosomes will be in middle layer, generative cell and male gametes?
- (1) 14, 28, 14
 - (2) 28, 14, 14
 - (3) 14, 14, 28
 - (4) 28, 28, 14
108. Read the statements (A–D) and choose the options having all the correct statements :-
- (A) A typical angiospermic anther is bilobed with each lobe having two theca and tetragonal in outline.
- (B) The number and length of stamens are variable in flowers of different species.
- (C) Endothecium is present in between middle layer and tapetum.
- (D) Flower is morphological and embryological marvel and considered as a chief site of sexual reproduction.
- (1) A, B, C, D
 - (2) A, B, D
 - (3) B, C, D
 - (4) A, B, C

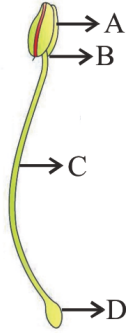
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PHASE - MED

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109. Identify the A, B, C, D in given diagram :-



	A	B	C	D
(1)	Anther	Proximal end	Filament	Distal end
(2)	Proximal end	Filament	Distal end	Anther
(3)	Anther	Distal end	Filament	Proximal end
(4)	Filament	Proximal end	Anther	Distal end

110. How many plant structures in the list below are haploid in nature.

Antipodals, pollen grain, Tapetum, embryo sac, ovule, synergids, nucellus, microspore, megaspore.

- (1) 6 (2) 7 (3) 5 (4) 8

111. Match the column-I with column-II and choose the correct answer :-

Column-I		Column-II	
(a)	Microsporangium	(i)	Pollen grain
(b)	Male gametophyte	(ii)	Germinating pollen grain
(c)	Microsporophyll	(iii)	Pollen-sac
(d)	Microspore	(iv)	Stamen
(e)	Micro-gamete	(v)	Male gamete

Option :

(1)	a-iv	b-ii	c-iii	d-i	e-v
(2)	a-iii	b-i	c-v	d-ii	e-iv
(3)	a-iii	b-ii	c-iv	d-i	e-v
(4)	a-i	b-ii	c-iii	d-v	e-iv

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PHASE - MED

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112. Match Column-I and Column-II

	Column-I		Column-II
A	Chlamydomonas	I	Gemmules
B	Sponge	II	Zoospores
C	Hydra	III	Conidia
D	Penicillium	IV	Buds

(1) A – II, B – IV, C – I, D – III

(2) A – IV, B – III, C – II, D – I

(3) A – II, B – I, C – IV, D – III

(4) A – I, B – III, C – IV, D – II

113. In seed plants the non-motile male gametes are carried to female gamete by :

- (1) Water (2) Insects
(3) Wind (4) Pollen tube

114. External fertilization can be observed in :-

- (1) Amphibians (2) Fishes
(3) Majority of algae (4) All of the above

115. Which of the following is/are considered as pre-fertilization events ?

- (1) Gametogenesis (2) Gamete transfer
(3) Embryogenesis (4) Both (1) and (2)

116. Parthenogenesis can be observed in -

- (1) Honeybees (2) Rotifers
(3) Some lizards (4) All of the above

117. In some plants both male and female flowers may be present on same individual (_A_) or on separate individuals (_B_)

- (1) A = Heterothallic, B = Homothallic
(2) A = Monoecious, B = Dioecious
(3) A = Dioecious, B = Monoecious
(4) A = Staminate, B = Pistillate.

118. The coconut water and the edible part of coconut are equivalent to :-

- (1) Mesocarp (2) Endocarp
(3) Endosperm (4) Embryo

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119. The arrangement of the nuclei in a normal embryo sac in the angiospermic plants is :-

- (1) 2 + 1 + 2 (2) 3 + 2 + 3
(3) 2 + 3 + 3 (4) 3 + 3 + 1

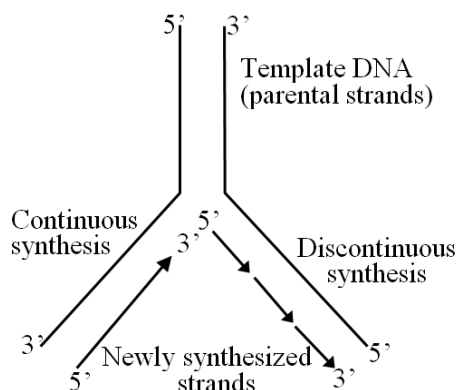
120. Of a gene, if both strands of DNA are transcribed then the two RNA molecules formed must be :-

- (1) Parallel and identical
(2) Antiparallel and complementary
(3) Parallel and complementary
(4) Antiparallel and Identical

121. Histones are rich in

- (1) alanine and glycine
(2) lysine and arginine
(3) histidine and serine
(4) cysteine and tyrosine

122.



Okazaki fragments are joined by :-

- (1) DNA ligase (2) DNA polymerase
(3) Helicase (4) Primase

123. Central dogma in molecular biology was given by :-

- (1) Erwin Chargaff (2) Rosalind Franklin
(3) George Gamow (4) Francis Crick

124. The unequivocal proof that DNA is genetic material came from the experiments of :-

- (1) Messelson and stahl
(2) Griffith
(3) Taylor
(4) Hershey and Chase

125. Replication does not initiate randomly at any place in DNA. It is a definite region where the replication originates such regions is called

- (1) Replication fork
(2) Origin of replication
(3) Promoter site
(4) Pallindromic Sequence

126. A molecule that can act as genetic material must not have following properties :-

- (1) It should be chemically and structurally highly reactive
(2) It should be able to generate its replica
(3) It should provide scope for slow changes
(4) It should be able to express in form of Mendelian characters

127. Which of the following act as substrate as well as provide energy for polymerisation during DNA replication?

- (1) ATP (2) GTP
(3) NTP (4) d NTP

128. Which of the following steps in transcription is catalysed by RNA polymerase ?

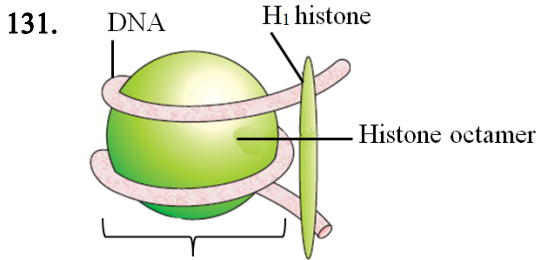
- (1) Initiation (2) Elongation
(3) Termination (4) All of these

129. Which of the following RNA act as enzyme for the formation of peptide bond in bacteria ?

- (1) 28s rRNA (2) 23s rRNA
(3) 16s rRNA (4) 18s rRNA

130. The strand of DNA, which does not code for anything is referred as ?

- (1) Template strand
(2) Antisense strand
(3) Non-coding strand
(4) Coding strand



Identify this figure :-

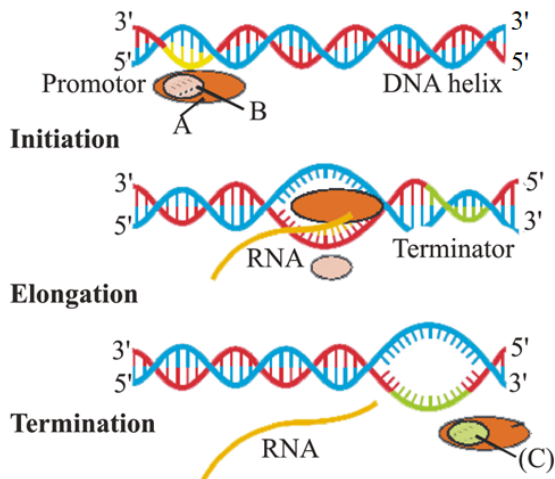
- (1) Spliceosome (2) Nucleosome
(3) Informosome (4) Polysome

132. Find the correct match in column-I and II :-

	Column-I		Column-II
(a)	Exon	(i)	Site for binding of RNA Polymerase
(b)	Capping	(ii)	Coding sequence
(c)	Tailing	(iii)	Methyl guanosine triphosphate
(d)	Promotor	(iv)	Adenylate residues

- (1) a - ii, b - iii, c - iv, d - i
(2) a - iv, b - i, c - ii, d - iii
(3) a - i, b - iv, c - iii, d - ii
(4) a - iii, b - ii, c - i, d - iv

133. In the given picture correctly identify A, B and C:-



- (1) A-Sigma factor, B-Rho factor, C-Stop codon
(2) A-RNA polymerase, B-Sigma factor, C-Stop codon
(3) A-Rho factor, B-RNA polymerase, C-Terminator
(4) A-RNA polymerase, B-Sigma factor, C-Rho

134. Non-sense or stop codons are-

- (1) AUG, GUG, CUG
(2) UAA, UAG, UGA
(3) AUU, ACU, ACC
(4) UGG, UGA, UGC

135. The given sequence when translated must form a polypeptide of how many amino acids ?

5'CUCAAUCCAUGCCCUUGGUUAGCGA-3'

- (1) Eight (2) Nine
(3) Six (4) Five

SECTION-B (BIOLOGY-I)

136. Typical angiosperm anther is :-

- (A) Dithecous (B) Bilobed
(C) Monothealous (D) Unilobed

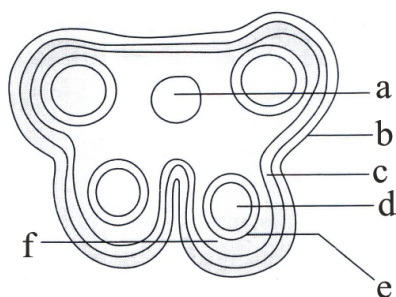
- (1) Both A and B are correct
(2) Both B and C are correct
(3) Only B is correct
(4) Only A is correct

137. Which of the following pair of statement are false ?

- a. Pollen grains of *Parthenium* and *Ambrosia* are not aeroallergen.
b. Pollen robbers consume pollen without bringing pollination..
c. Sporopollenin is the inorganic material and considered as hardest material of biological world.
d. Xenogamous flowers may be bisexual or unisexual.

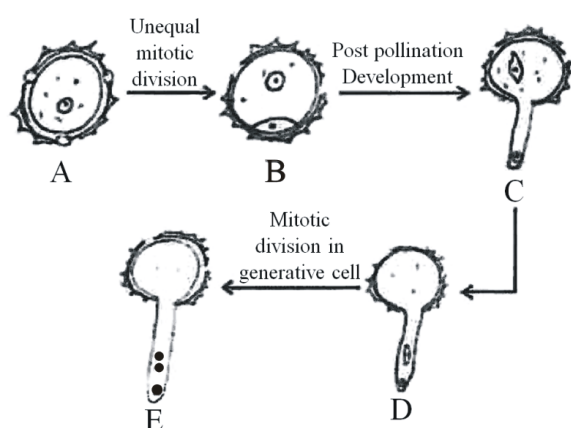
- (1) a and b
(2) b and c
(3) c and d
(4) a and c

138. Recognise the figure and find out the correct matching :-



- (1) a-epidermis, b-endothecium, c-middle layers, d-tapetum, e-sporogenous tissue, f-connective
- (2) b-epidermis, c-endothecium, d-middle layers, e-tapetum, f-sporogenous tissue, a-connective
- (3) b-epidermis, c-endothecium, f-middle layers, e-tapetum, a-sporogenous tissue, d-connective
- (4) b-epidermis, c-endothecium, f-middle layers, e-tapetum, d-sporogenous tissue, a-connective

139. The figure given below represents the various steps of microgametogenesis. Select the option giving correct identification together with what it represents ?



- (1) (C) Represents mature male gametophyte
- (2) (A) Represents mature pollen grain
- (3) (A) Represents mature male gametophyte
- (4) (E) Represents mature male gametophyte

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140. Which of the following is not correct ?

- (1) The annual and biennial plants show clear cut vegetative, reproductive and senescent phases
- (2) Bamboo flower generally after 50-100 years produce large numbers of fruits and die.
- (3) *Strobilanthus kunthiana* (Neelakuranji) flowers in every 02 years from the time of plantation.
- (4) The reproductive phase is of variable duration in different organisms

141. In diploid organisms, specialised cells _____ undergo meiosis.

- (1) Zygote
- (2) Meiocytes
- (3) Gametophyte
- (4) Somatic cell.

142. Gametes are haploid though the parent plant body from which they are arise -

- (1) Haploid always
- (2) Diploid always
- (3) Either haploid or diploid
- (4) Polyploid always

143. How many from the examples given below are unisexual and bisexual animals respectively ?

- (A) Cockroach
- (B) Earthworm
- (C) sponge
- (D) Tapeworm
- (E) Leech

- (1) D and A
- (2) C and B
- (3) A and D
- (4) B and C

144. Identify monoecious and dioecious plants respectively from the given examples below :-

- (A) Papaya
- (B) Cucurbits
- (C) Date palm
- (D) Coconut

- (1) B and B
- (2) A and C
- (3) C and A
- (4) D and zero

145. What would be the number of chromosome of the aleurone cells of a plant with 32 chromosomes in its synergid cell ?

- (1) 96 (2) 64 (3) 8 (4) 16

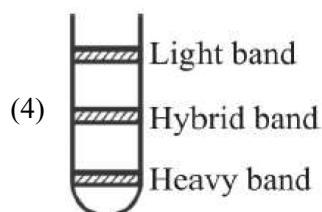
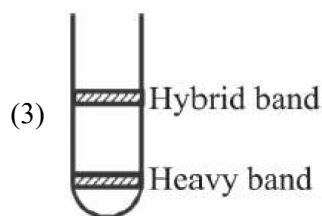
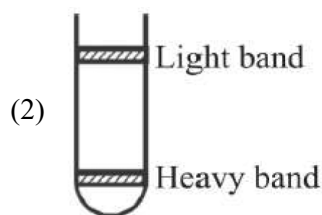
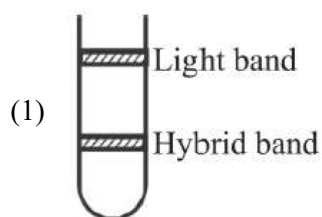
146. Male gametophyte with least number of cell is present in :-

- (1) Fern (2) Cycas
(3) Moss (4) Mango

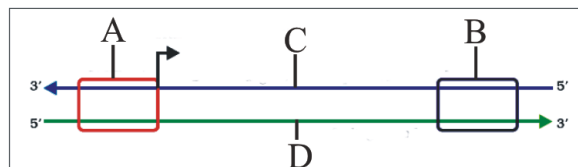
147. In the year 1928, Frederick Griffith showed that the hereditary material can pass from one type of bacteria into another through a process called :-

- (1) transduction (2) conjugation
(3) recombination (4) transformation

148. Light DNA molecule is replicated in a heavy medium. After the four successive replication of the DNA, 16 DNA molecule are formed. Which one of the following is true for these DNA:-



149. Given diagram represent the component of a transcription unit selection the correct answer regarding it.

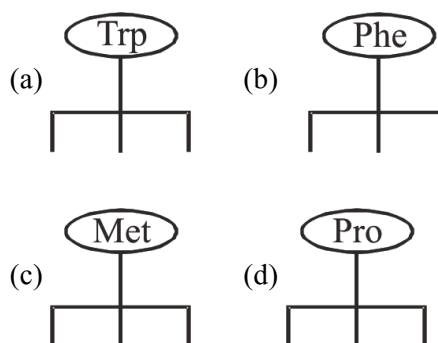


	A	B	C	D
(1)	Terminator	Promoter	Template strand	Coding strand
(2)	Promoter	Terminator	Coding strand	Template strand
(3)	Promoter	Terminator	Template strand	Coding strand
(4)	Terminator	Promoter	Coding strand	Template strand

150. mRNA sequence is given :-

5' AUG UUC CCG UGG 3'

What will be the correct sequence of charged tRNAs ?



- (1) a, b, c, d
(2) d, c, b, a
(3) c, a, d, b
(4) c, b, d, a

Topic : DIGESTION AND ABSORPTION, MOLECULAR BASIS OF INHERITANCE UPTO TRANSLATION, TAXONOMY TO BIOLOGICAL CLASSIFICATION AND MONERA [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM

SECTION-A (BIOLOGY-II)

151. In translation, initiator-tRNA brings first amino acid, which is :-

- (1) Glycine
- (2) Methionine
- (3) Tryptophan
- (4) Valine

152. In translation which of the following step utilize energy in the form of ATP ?

- (1) charging of t-RNA
- (2) Amino acid activation
- (3) Translocation of ribosome
- (4) Elongation of polypeptide chain

153. Which RNA is Ribozyme here ?

- (1) 23 s rRNA
- (2) 28 s rRNA
- (3) Sn RNA
- (4) All of the above

154. Given below are two statements :

Statement I : DNA polymerases catalyse polymerisation only in one direction, that is $5' \rightarrow 3'$

Statement II : During replication of DNA, on one strand the replication is continuous while on other strand it is discontinuous.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) Both **Statement I** and **Statement II** are correct.

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PHASE - MED

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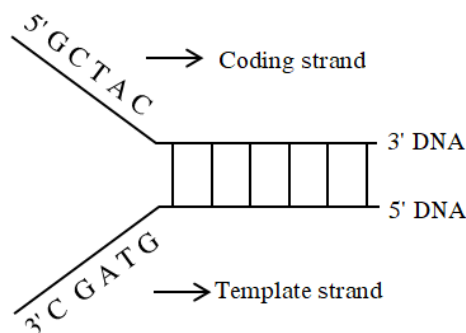
155. Amino acid attaches to the tRNA at it's $\rightarrow$

- (1) Acceptor arm
- (2) Anticodon loop
- (3) DHU loop
- (4) T Ψ C loop

156. In Eukaryotes, mRNA constitutes by all of the following **EXCEPT** :

- (1) Cap at 5' end
- (2) Poly A Tail at 3' end
- (3) Exons
- (4) Introns

157.



What will be the sequence of RNA ?

- (1) 5' GCUAC 3'
- (2) 5' CGAUG 3'
- (3) 5' GCTAC 3'
- (4) 5' CGATG 3'

158. In protein synthesis, ___a___ acts as template, ___b___ brings amino acids and ___c___ play structural and catalytic role

	a	b	c
(1)	tRNA	rRNA	mRNA
(2)	mRNA	tRNA	rRNA
(3)	rRNA	mRNA	tRNA
(4)	mRNA	rRNA	tRNA

159. Choose an incorrect statement based on splicing process :-

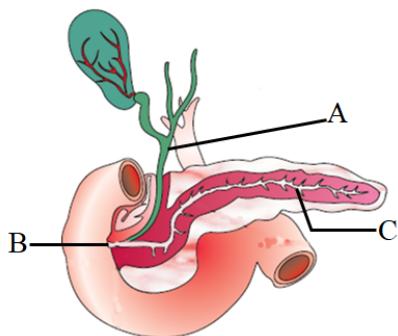
- (1) Process of splicing represents the dominance of RNA world.
- (2) Presence of introns is reminiscent of antiquity.
- (3) Split - gene arrangements represent an advanced feature of genome.
- (4) It is RNA - dependent process.

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160. Ribosome consists of ____A____ RNAs and ____B____ different proteins.

	A	B
(1)	Structural	80
(2)	Adapter	80
(3)	Messenger	80
(4)	Catalytic	70

161.



Identified A,B,C.

- (1) A-common Bile duct, B-Sphincter of oddi, C pancreatic duct
- (2) A-Hepatic duct, B-Hepato-pancreatic duct, C-pancreatic duct
- (3) A-C.B.D, B-Hepato-pancreatic duct, C-Pancreatic duct
- (4) A-Cystic duct, B-Sphincter of oddi, C-Pancreatic duct
162. **Assertion :** Pepsinogen is a pro enzyme.
Reason : It is secreted in an inactive state and on exposure to HCl gets converted into pepsin.
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.
163. Which salivary glands is present in cheeks ?
- (1) Sub-maxillary (2) Sub-mandibular
- (3) Parotid (4) Sub-Linguals

TG: @Chalnaayaaar

164. As a doctor, you diagnose a patient who suffer from a disease. On diagnosis you find following symptoms. Yellow color of eyes, skin and nails and liver is not function properly. This condition is likely due to which of the following ?

- (1) Deposition of Bile
- (2) Deposition of Bile pigments
- (3) Deposition of Bile salt like Bilirubin
- (4) All of the above

165. Select the correct match for enzyme and function

- (1) Ptyalin → 70% digestion of starch
- (2) Gastric Lipase → Helps in emulsification of fat
- (3) Lysozyme → Kill bacteria and viruses
- (4) Chymosin → Helps is digestion of milk protein

166. Digestive enzyme which act upon alkaline pH are / is :-

- (1) Bilirubin (2) Biliverdin
- (3) HCl (4) Steapsin

167. Which of the following is not endodermal in origin-

- (1) Oesophagus (2) Liver
- (3) Teeth (4) Small intestine

168. Select incorrect option for function of liver :-

- (1) Bile formation (2) Erythropoiesis in adult
- (3) Glycogenolysis (4) Deamination

169. Select the incorrect match w.r.t. Digestive system-

- (1) Duodenum - Maximum digestion of food
- (2) Jejunum - Maximum absorption of food
- (3) Large Intestine - Digestion of food complete
- (4) Stomach - Store food for 4 to 5 hours

170. Select the correct option for widest part of alimentary canal-

- (1) 'J' shape bag like structure
- (2) Situated on left upper side of abdominal cavity
- (3) Thickest muscle layer present
- (4) All of these

171. Archaeobacteria differ from other eubacteria in having :-

- (1) Both do not have true nucleus
- (2) Composition of cell wall
- (3) Composition of cytoplasm
- (4) Composition of genetic material

172. Which of the following feature is not found in Monera Kingdom?

- (1) Structure very complex
- (2) Very complex behaviour
- (3) Most extensive metabolic diversity
- (4) Vast majority heterotroph

173. In Whittaker's five kingdom system of classification, eukaryotes are distributed among:-

- (1) Two kingdoms
- (2) Three kingdoms
- (3) Four kingdoms
- (4) All five kingdoms

174. Which of the following is not a part of cell envelope ?

- (1) Cell wall
- (2) Glycocalyx
- (3) Cell membrane
- (4) Nucleoid

175. Which of the following structure is responsible to attach the bacteria to host tissues and to rocks in streams ?

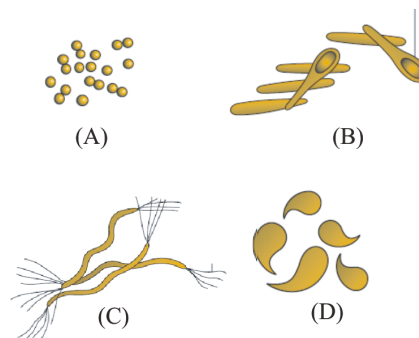
- (1) Pili
- (2) Flagella
- (3) Capsule
- (4) Fimbriae

TG: @Chalnaayaaar

176. Which of the following structure of bacteria role in motility.

- (1) Pilli
- (2) Flagella
- (3) Fimbriae
- (4) Basal body

177. Identify the following shapes (A, B, C and D) of bacteria :-



	A	B	C	D
(1)	Spirilla	Bacilli	Cocci	Vibrio
(2)	Cocci	Bacilli	Spirilla	Vibrio
(3)	Vibrio	Spirilla	Bacilli	Cocci
(4)	Bacilli	Cocci	Vibrio	Spirilla

178. Find the incorrect statements from the following:-

- (A) Bacterial structure is very simple but very complex in behaviour
- (B) Bacteria show the most extensive metabolic diversity
- (C) The vast majority of bacteria are autotrophs and a few are heterotrophs
- (D) Bacterial cell wall is made up of lipoprotein

- (1) Only A & B
- (2) Only C & D
- (3) Only A & D
- (4) Only B & C

179. The term "Taxonomy" given by

- (1) Bentham & Hooker
- (2) A.P. de Candolle
- (3) Theophrastus
- (4) Linnaeus

SECTION-B (BIOLOGY-II)

180. Binomial system of nomenclature was given by :-

- (1) Lamarck
- (2) Theophrastus
- (3) Linnaeus
- (4) Hutchinson

181. Ascending order of rank of taxa are :

- (1) Genus, species, order, family
- (2) Species, genus, family, order
- (3) Species, genus, order, family
- (4) Family, order, genus, species

182. The word ending with 'aceae' indicates

- (1) Genera
- (2) Class
- (3) Family
- (4) Order

183. Taxonomic hierarchy refer to :

- (1) A group of senior taxonomist who decide the nomenclature of plant and animal.
- (2) Stepwise arrangement of all categories for classification of plant and animal.
- (3) A list of botanist or zoologist who have work on taxonomy
- (4) Classification of a species based on fossil record.

184. 'Taxa' differ from 'taxon' due to

- (1) This being a higher taxonomic category than taxon
- (2) Lower taxonomic categories than taxa
- (3) This being the plural of taxon
- (4) This being the singular of taxon

185. Linnaeus 'classification' is based on :

- (1) Plant morphology
- (2) Type of stem
- (3) Number of leaves
- (4) Number of stamen

TG: @Chalnaayaaar

186. **Assertion** : The process of translation requires transfer of genetic information from a polymer of nucleotides to form a polymer of amino acids.

Reason : This process take place in cytoplasm of cell.

- (1) Both assertion & reason are true & the reason is a **correct** explanation of the assertion.
- (2) Both assertion & reason are true but reason is **not** a correct explanation of the assertion
- (3) Assertion is **true** but the reason is **false**
- (4) Both assertion & reason are **false**.

187. Choose correct statements about genetic codes ?

- (i) Un-ambiguous
- (ii) Some amino acids are coded by more than one codon
- (iii) If any nucleotide is added or deleted, then from this point, genetic codes read differently
- (iv) Nearly universal

- (1) i, ii
- (2) ii, iii, iv
- (3) i, ii, iii
- (4) i, ii, iii, iv

188. Name the enzyme that facilitates opening of DNA helix during transcription.

- (1) RNA polymerase
- (2) DNA ligase
- (3) DNA helicase
- (4) DNA polymerase

189. If mRNA has 132 bases, from which 100th base is removed, then how many codons will be changed ?

- (1) 11
- (2) 33
- (3) 1
- (4) 10

190. If a codon in mRNA is 5'AUG3', then anticodon in tRNA will be "

- (1) 5'UAC3'
- (2) 5'CAU3'
- (3) 5'AUG3'
- (4) 5'GUA3'

191. **Statement-I** : Brunner's gland is present in sub-mucosa of Duodenum.

Statement-II : Intrinsic factor essential for absorption of vit. B₁₂

- (1) Statement-I is correct but statement-II is incorrect
- (2) Both statement are correct
- (3) Both statement are incorrect
- (4) Statement-I is incorrect but statement-II is correct

192. Read the following statement -

- (A) Stomach store food for 4-5 hours.
- (B) Food mix with gastric juice at 1.8 pH known as bolus.
- (C) Bolus contain enzymes that act on 6.8 pH.
- (D) Enterokinase convert trypsinogen into trypsin.
- (E) Trypsin digest peptones into proteoses.

Which of the following option is correct for given statement ?

- (1) A, B, D are false and C, E true
- (2) A, C, D are true and B, E are false
- (3) A, C, D are false and B, E are true
- (4) B, C, D are true and A, E are false

193. In the given components/substances, how many are NOT part of bile juice ?

Bilirubin, Biliverdin, Cholesterol, Phospholipids, Maltase, Nucleases, HCl, Intrinsic factor.

- (1) 4
- (2) 3
- (3) 2
- (4) 6

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194. Fill up the blanks.

- (I) Three kingdom system of classification was proposed by ...(A)..
- (II) Four kingdom system of classification was proposed by(B)...
- (III) ...(C)... includes prokaryotic microorganism like ...(D)...

Identify A-D to complete the given statement:-

- (1) A-Aristotle, B-Whittaker, C-Protista, D-Viroids
- (2) A-Ernst Haeckel, B-Copeland, C-Monera, D-bacteria
- (3) A-Linnaeus, B-Stainier, C-Fungi, D-virus
- (4) A-Whittaker, B-Von Niel, C-Protozoa, D-prions

195. In the five kingdom classification which kingdom occupies intermediate position from phylogenetic point of view :-

- (1) Monera
- (2) Mycota
- (3) Plantae
- (4) Protista

196. Extra chromosomal, small, circular DNA in prokaryotes is termed as :-

- (1) Chromatophore
- (2) Plasmids
- (3) Polysome
- (4) Genophore

197. Find out the correct match from the following table :-

	Coloumn I	Coloumn II	Coloumn III
(i)	Mesosomes	Oxidative Enzyme	Cell respiration
(ii)	Capsule	Loose nature	Protein Synthesis
(iii)	Flagella	Flagellin Protein	Help in genetic recombination

- (1) (i) and (ii)
- (2) (ii) and (iii)
- (3) Only (iii)
- (4) Only (i)

198. Who gave natural system of classification

- (1) Takhtajan
- (2) Bentham & Hooker
- (3) Lamarck
- (4) Darwin

199. Classification based on chromosome number is :

- (1) Cytotaxonomy
- (2) Numerical taxonomy
- (3) Chemo taxonomy
- (4) Both (1) and (3)

200. Which one of the following is not a correct statement.

- (1) Botanical garden have collection of living plants for reference
- (2) A museum has collection of photographs of plant and animals
- (3) Key is a taxonomic aid for identification of specimens
- (4) Herbarium house dries, pressed and preserved plant specimens

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MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
06 13-08-2023	SYLLABUS COVERED FROM 17-07-2023 TO 12-08-2023	P	RAY OPTICS & OPTICAL INSTRUMENTS
		C	NOMENCLATURE, ISOMERISM
		B	ECOLOGY

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